

Environmental and Development Economics

Module 2 - How does development affect the environment?

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Lecture 2

How does development affect the environment?

Housekeeping

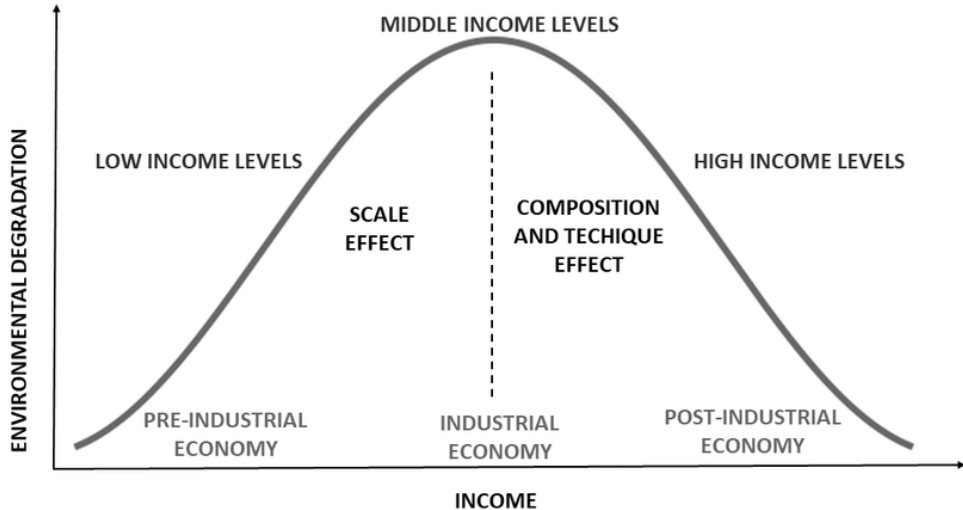
- ▶ Paper summaries are random **without** replacement (sorry!)
- ▶ All assignment instructions + rubrics posted
- ▶ Research proposal (first draft Oct 5th)
 - ▶ Start thinking about idea + come talk to me
 - ▶ DO NOT re-use paper from another class
- ▶ First draft expectations
 - ▶ **AT LEAST** 1 page on each section
- ▶ See VoxDevLit on Deforestation

- ▶ **Guiding question:** how does economic development affect the environment?
- ▶ Descriptive overview: EKC
- ▶ Channel I: Income effects (today)
 - ▶ **Changes in consumption**
- ▶ Channel II: Technology and Infrastructure (next time)
 - ▶ Agricultural productivity
 - ▶ Infrastructure
- ▶ Channel III: Institutions
 - ▶ Later in the course

How does economic development affect the environment?

- ▶ There is no one answer
- ▶ Choices that maximize economic growth often degrade environment
- ▶ But development expands our choice set (e.g. clean energy)
- ▶ Development also lowers $u'(c)$ (model from last week)
- ▶ Hence: the Environmental Kuznets Curve (EKC)

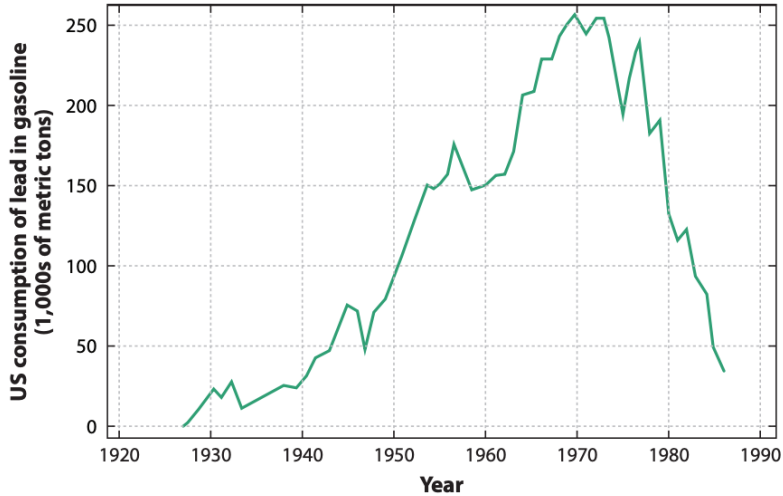
Environmental Kuznets Curve



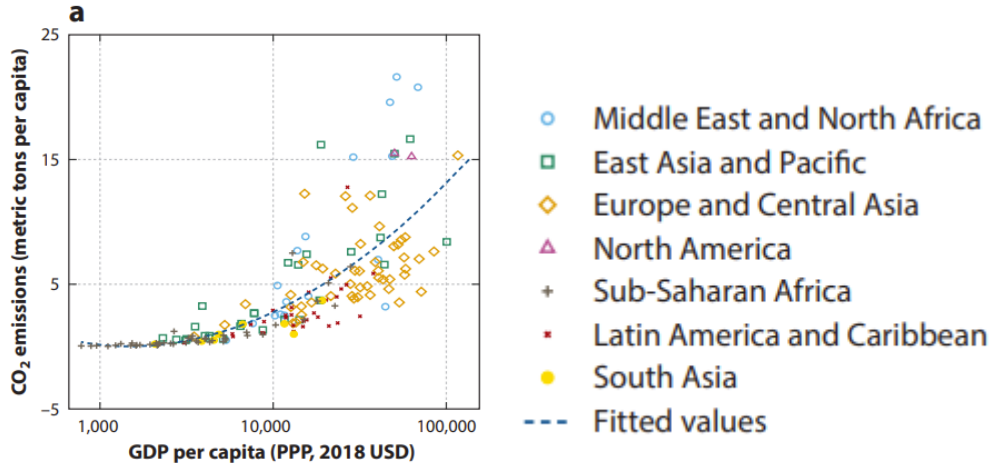
Discussion: Do you believe EKC?

- ▶ Is it true for all pollutants?
- ▶ Causality?
- ▶ How to find the tipping point?

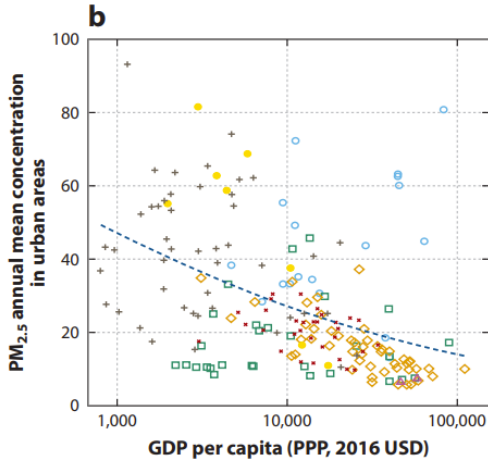
GDP and Lead in Gasoline



GDP and CO2

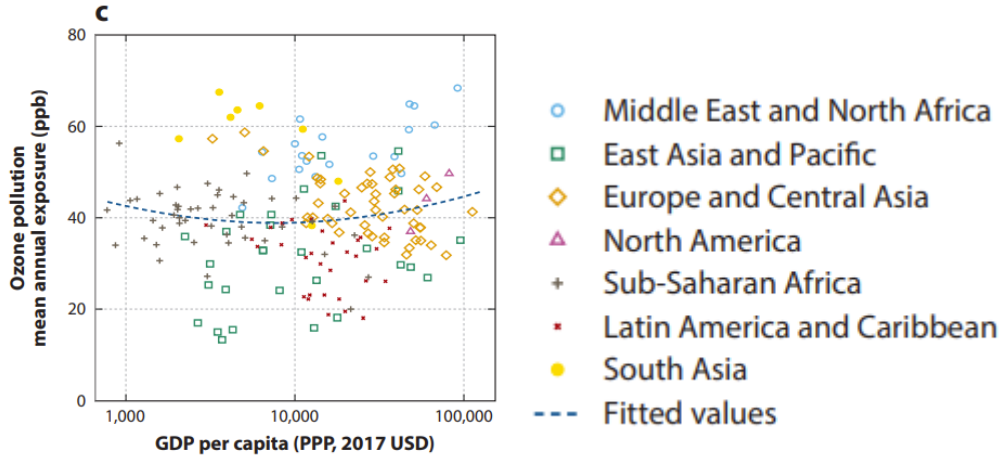


GDP and PM2.5



- Middle East and North Africa
- East Asia and Pacific
- ◇ Europe and Central Asia
- △ North America
- + Sub-Saharan Africa
- × Latin America and Caribbean
- South Asia
- Fitted values

GDP and Ozone



My view

- ▶ Macro correlations give limited answer
 - ▶ At best, we can learn about cross-country relationships
- ▶ Not the right question
 - ▶ if we want to know how development affects environment, we need: 1) causality, 2) micro data, and 3) **a better definition of development**
 - ▶ e.g. manufacturing ↑ pollution, but services may not
- ▶ A better question:
 - ▶ Which component of dev't affects which enviro outcome, through what mechanism?
- ▶ Narrow, micro/empirical papers more helpful
 - ▶ even if they don't find consensus

Discussion: What is the ideal (quasi-) experiment?

- ▶ Can we randomize GDP?
- ▶ What variation in “development” can approximate the experiment?
 - ▶ what proxies development? (see ookla speed test)
 - ▶ what does the proxy miss?
 - ▶ what (non-random) variation isolates causal relationship?

Discussion: What about mechanisms?

- ▶ **Why and how** does economic development affect the environment?
 - ▶ Ex: energy, diet, self-protection, capital, institutions
- ▶ Theory is your friend: especially total derivatives. . .
 - ▶ Otherwise generate testable predictions
- ▶ Heterogeneity is your friend
 - ▶ But have a reason **why** you choose one channel over another

Question: What is the effect of income on environmental quality?

- ▶ Poverty alleviation may $\uparrow$ demand for resource-intensive good
- ▶ Or, it may $\uparrow$ demand for conservation
- ▶ **This paper:** Exploit RDD eligibility to study impact of CCT on forests

Why can't we regress forests on income?

- ▶ Income correlated with other things that affect environment
- ▶ Like what?
- ▶ If we control for OVB, will panel fixed effects work?

Mexico's Oportunidades Program

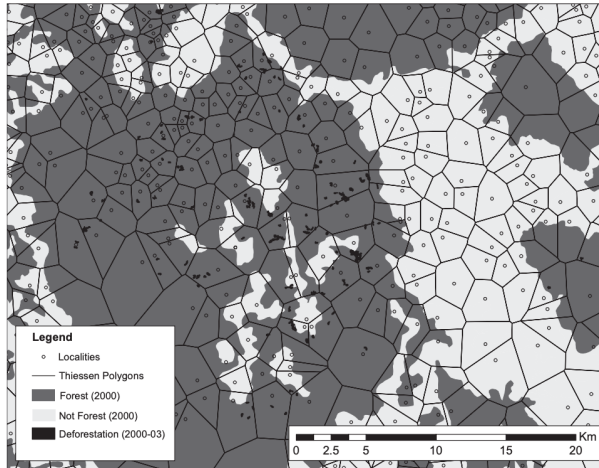
- ▶ **Goal:** Ambitious program to build human capital
- ▶ **Scope:** 4 million households treated
 - ▶ \$2.6 billion/year (half of anti-poverty budget)
 - ▶ Transfers are 1/3 of total income
 - ▶ Represents shift to new income path
- ▶ **Design:** Randomized pilot + non-randomized rollout
 - ▶ 1997: randomized pilot at village level
 - ▶ 1998-2000: rapid rural rollout based on **marginality** (not random)
 - ▶ Post-2000: new villages enroll, but slower
 - ▶ Within villages, household selected based on marginality index

Results Preview: conditional cash transfer increases deforestation

- ▶ **Main variables:** village lat/lon, eligibility index, forest cover (satellite)
- ▶ **Sample size:** 105,749 villages
- ▶ **Unit of analysis:** only village centroids provided
 - ▶ How do we estimate village-level impacts? Thiessen polygons
 - ▶ Area around a point where every location closer to the point than to all others
 - ▶ Problems?

Thiessen/Voronoi Polygons

FIGURE 2.—ILLUSTRATION OF LOCALITY BOUNDARIES DEFINED USING THIESSEN POLYGONS



Eligibility cutoff: -1.2

FIGURE 3.—ENTIRE SAMPLE MINUS OBSERVATIONS WITH INDEX > 3

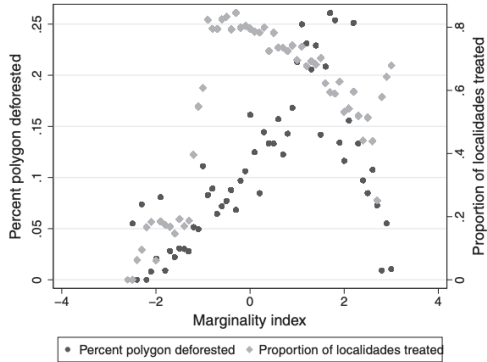
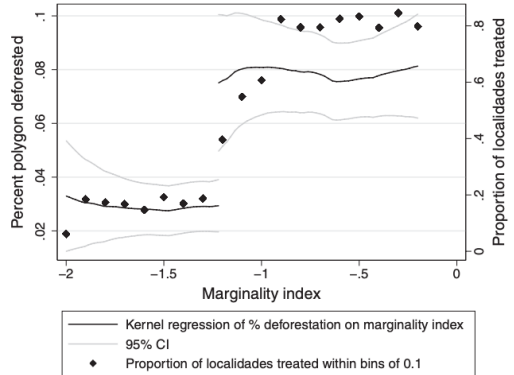


FIGURE 4.—KERNEL ESTIMATION OF DEFORESTATION ON MARGINALITY INDEX—RESTRICTED SAMPLE



Empirical Strategy: OLS Equation

$$\Delta f_i = \alpha + \delta T_i + \beta' X_i + \epsilon_i$$

- ▶ where Δf_i is % deforested in polygon i over 2000-2003
- ▶ $T_i = 1$ if the locality enrolled by 2003
- ▶ X_i are covariates, including poverty
- ▶ But T_i is not random!

- ▶ **Solution:** Regression Discontinuity Design
- ▶ If discontinuity is sharp, replace T_i with $1[E_i > -1.2]$
- ▶ RDD: Sample window $-2 < I < -0.2$

RDD Equation

- ▶ Discontinuity is fuzzy, not sharp
 - ▶ How do we know? Two reasons
- ▶ Solution: Use cutoff to instrument treatment probability
- ▶ Fuzzy RDD/IV:

$$\Delta f_i = \alpha + \delta T_i + \gamma l_i + \beta' X_i + \epsilon_i$$

$$\Delta T_i = \omega + \tau_1 E_i + \tau_2 E_i l_i + \tau_3 M_i + \tau_4 M_i l_i + \mu l_i + \Gamma' X_i + \epsilon_i$$

where T_i is enrollment dummy, E_i is eligibility cutoff dummy, l_i is index, M_i is dummy for region where enrollment increases rapidly

Identification Assumptions

Results: OLS and Sharp RDD

	Tobit			OLS	
	% Polygon Deforested			Deforestation (0/1)	% Deforested (If 1)
	(1)	(2)	(3)	(4)	(5)
Eligible	.383 (.181)**	.549 (.295)*	.370 (.217)*	.013 (.008)*	.387 (.190)**
Marginality index	.523 (.041)***	.753 (.077)***	.219 (.189)	.031 (.003)***	.069 (.075)
Index ²		.069 (.072)		.002 (.003)	.060 (.075)
Index ³		-.100 (.037)***		-.004 (.001)***	-.022 (.025)
Index ⁴		-.002 (.015)		-.0001 (.0005)	-.012 (.013)
Baseline area in forest, 2000	-.0004 (.001)	-.0005 (.001)	.004 (.002)**	.0006 (.0001)***	.005 (.001)***
Ln(polygon area)	.947 (.042)***	.954 (.042)***	.728 (.068)***	.046 (.002)***	-.993 (.062)***
Ln(total population in 1995)	.142 (.024)***	.144 (.024)***	.036 (.034)	.010 (.001)***	-.040 (.025)
Ln(slope)	-.052 (.005)***	-.053 (.005)***	-.009 (.010)	-.003 (.0002)***	-.029 (.006)***
Ln(road density)	-.059 (.026)**	-.056 (.026)**	.025 (.053)	-.004 (.001)***	-.010 (.027)
Observations	58,587	58,587	15,758	58,587	5,545
Ecoregion controls	Yes	Yes	Yes	Yes	Yes

Results: Fuzzy RDD / Instrumental Variables

- Very strong first stage: eligibility $\uparrow$ probability of enrollment ($\tau_1 = 0.8$)

	IV Tobit				IV OLS	
	Full Estimation Sample			Restricted Sample	Deforestation (0/1)	% Deforested (If 1)
	(1)	(2)	(3)	(4)	(5)	(6)
Treated	.584 (.280)**	1.293 (.715)*		1.038 (.609)*	.031 (.019)*	1.264 (.680)*
Proportion treated			3.453 (1.870)*			
Marginality index	.521 (.042)***	.641 (.106)***	.244 (.298)	-.072 (.339)	.028 (.003)***	-.005 (.101)
Index ²		.177 (.116)	.391 (.221)*		.004 (.004)	.162 (.119)
Index ³		-.091 (.035)***	-.053 (.031)*		-.003 (.001)***	-.036 (.030)
Index ⁴		-.010 (.015)	-.037 (.022)*		-.0003 (.0005)	-.019 (.014)
Baseline area in forest, 2000	-.0005 (.001)	-.0008 (.001)	-.001 (.001)	.003 (.002)**	.0006 (.0001)***	.004 (.001)***
Ln(polygon area)	.963 (.043)***	.990 (.047)***	1.075 (.079)***	.756 (.070)***	.047 (.002)***	-.948 (.065)***
Ln(total population in 1995)	.055 (.051)	-.056 (.116)	-.305 (.245)	-.097 (.086)	.005 (.003)	-.262 (.120)**
Ln(slope)	-.054 (.005)***	-.057 (.006)***	-.064 (.008)***	-.012 (.010)	-.003 (.0002)***	-.033 (.007)***
Ln(road density)	-.075 (.027)***	-.092 (.033)***	-.119 (.043)***	.016 (.054)	-.005 (.001)***	-.049 (.036)
Observations	58,587	58,587	58,587	15,758	58,587	5,545
Ecoregion controls	Yes	Yes	Yes	Yes	Yes	Yes

- ▶ RD results show that CCT increases deforestation
- ▶ Why? What changes at the household level?
- ▶ Approach 1: Use experimental sample (Progresa) with household survey data
 - ▶ Progresa had randomized rollout at village level for first 3 yrs
- ▶ Approach 2: Heterogeneity by road density to study role of market access

Mechanisms: Consumption channel

- ▶ Before (1997-1998) and after (2000) data on consumption
- ▶ 506 villages, 320 treated (early intervention), 186 control (delayed intervention)
- ▶ Treatment at the village level

Difference in differences:

$$y_{it} = \gamma_0 + \gamma_1 Treat_i + \gamma_2 Post_t + \gamma_3 Treat_i \times Post_t + \epsilon_{it}$$

- ▶ y_{it} = consumption
- ▶ $Treat_i = 1$ if household i in treated locality
- ▶ cluster at locality level bc randomization was at locality level
- ▶ How do we interpret γ_3 ?

Hypothesis: $\gamma_3 > 0$

Mechanisms: Market channel

- ▶ Demand shock must be met by supply, which drives land use change
 - ▶ e.g. $\uparrow$ consumption of milk met by $\uparrow$ in grazing land
- ▶ But γ_3 captures partial equilibrium
 - ▶ Part of demand shock supplied locally. What about rest?
- ▶ Hypothesis: If consumption is driving deforestation
 - ▶ Then effect should be larger when infrastructure quality is low
 - ▶ Low market access: demand met by local supply
 - ▶ High market access: demand shock propagates across markets

Triple Differences:

Heterogeneity by infrastructure quality:

$$\begin{aligned} y_{it} = & \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 (Treat_i \times Post_t) \\ & + \beta_4 Road_i + \beta_5 (Road_i \times Treat_i) + \beta_6 (Road_i \times Post_t) \\ & + \beta_7 (Road_i \times Treat_i \times Post_t) + \epsilon_{it} \end{aligned}$$

► $Road_i$ = inverse road density (km of road w/n 10km of village/polygon)

Hypothesis: $\beta_7 > 0$

Results: Increase in consumption of land intensive goods

	Rooms in Home		Days Ate Beef		Days Drank Milk	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment effect	.014 (.033)	.017 (.035)	.114 (.030)***	.118 (.031)***	.337 (.081)***	.331 (.087)***
Treatment × inverse road density		−.034 (.148)		−.070 (.097)		.183 (.669)
Village chosen to receive Progresa	.0001 (.037)	.002 (.038)	−.025 (.029)	−.031 (.030)	−.133 (.111)	−.143 (.118)
Posttreatment year	.053 (.028)*	.049 (.029)*	−.137 (.024)***	−.138 (.025)***	−.655 (.061)***	−.664 (.065)***
Inverse of road density		.266 (.169)		−.156 (.069)**		.051 (.499)
Village × inverse road density		.043 (.236)		.102 (.140)		.232 (.682)
Posttreatment × inverse road density		.067 (.140)		.016 (.068)		.155 (.252)
Observations	23,318	23,318	33,128	33,128	33,128	33,128
Mean dependent	1.557		0.388		1.440	
Variable in baseline	(0.930)		(0.661)		(2.367)	

Higher beef and milk demand (land intensive products)

Demand-side impacts do not vary with market access (we didn't expect it to)

Results: No increase in local production

	Number of Plots		Log (1+ Total Hectares)		Number of Cows	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment effect	.030 (.039)	.031 (.040)	-.014 (.038)	-.015 (.039)	.092 (.057)	.036 (.057)
Treatment $\times$ inverse road density		-.107 (.210)		.142 (.223)		.936 (.522)*
Village chosen to receive Progresa	.014 (.056)	.037 (.057)	-.004 (.040)	.017 (.040)	-.004 (.087)	.058 (.085)
Posttreatment year	-.094 (.032)***	-.077 (.033)**	.312 (.033)***	.317 (.033)***	-.239 (.046)***	-.180 (.046)***
Inverse of road density		.833 (.161)***		.820 (.227)***		2.122 (.799)***
Village $\times$ inverse road density		-.263 (.317)		-.217 (.258)		-.760 (.872)
Posttreatment $\times$ inverse road density		-.275 (.149)*		-.235 (.128)*		-.982 (.402)**
Observations	45,087	45,087	32,631	32,631	34,248	34,248
Mean dependent	0.824		1.724		0.604	
Variable in baseline	(0.955)		(3.535)		(2.304)	

Supply-side impacts do not vary with market access

Results: Deforestation higher in places with poor market access

Dependent Variable	Low Density		Medium Density		High Density		Interactions	
	(%)	(0/1)	(%)	(0/1)	(%)	(0/1)	(%)	(0/1)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated	1.619 (.868)*	.075 (.037)**	.554 (.836)	.019 (.030)	1.818 (1.472)	.023 (.021)	.778 (.600)	.008 (.015)
Treated $\times$ low							1.041 (.582)*	.059 (.017)***
Low road density							-.550 (.400)	-.028 (.012)**
Observations	19,529	19,529	19,529	19,529	19,529	19,529	58,587	58,587

More deforestation in places with less market access (low road density)

Supports mechanism of increased demand for land-intensive goods

Thoughts?

- ▶ Do we believe the story about increased demand for land-intensive goods?
 - ▶ Less market access means less access to inputs
 - ▶ Underinvest in technology, leading to higher deforestation
 - ▶ Maybe this is a story about technology, not consumption
- ▶ Can you think of an alternate explanation?
- ▶ Other concerns?
- ▶ This all points to shortcomings of RCTs in general
 - ▶ we cannot answer why

- ▶ How does development affect the environment?
 - ▶ **Technology and Infrastructure**

Lecture 3

How does development affect the environment?

- ▶ I posted examples of assignments
- ▶ Research proposal first draft: Oct 5th
 - ▶ If you find idea this week, gives 2 weeks to write
- ▶ Class presentations: separate Canvas upload each week
- ▶ Replication project: start choosing a paper
- ▶ 9/25: Envirodevonomics Seminar with Sheetal Sekhri (UVA)
 - ▶ Cash Transfers and Adaptation to Extreme Heat: Evidence from Gig Workers in India

- ▶ Development increases access to capital
 - ▶ How do we think about environmental implications?
- ▶ Case I: Development increases agricultural productivity
 - ▶ Abman et al. (2023)
- ▶ Case II: Development spurs infrastructure development
 - ▶ Garg et al. (2023)

Technology and infrastructure

- ▶ As economies develop, technology and infrastructure improve
- ▶ Productivity boost means firms produce at lower cost
 - ▶ Outward shift in supply curve → bad for environment
 - ▶ But, technological progress reduces resource intensity → good for environment
- ▶ Most active areas of research:
 - ▶ agricultural productivity
 - ▶ infrastructure

Agricultural productivity and the environment

- ▶ **Boserup's hypothesis** : extensive margin
 - ▶ improvements in land productivity $\uparrow$ ag land value
 - ▶ leads to pressure on forests to clear land for ag
- ▶ **Borluag's hypothesis**: intensive margin
 - ▶ *Under factor market constraints*, $\rightarrow$ productivity spurs intensification
 - ▶ “spares” land for nature and puts less pressure on forests
- ▶ **Jevon's paradox**: general equilibrium
 - ▶ technology for increasing yield also lowers MC of producing given amount
 - ▶ Ag supply curve shifts out and may dominate gains in yield
 - ▶ Leads to net increase in land demand

Takeaway: this is an empirical question!

Conceptual framework

- ▶ Continuum of agents with heterogeneous outside (non-ag) options
 - ▶ Changes in returns to ag affect number of agents who select into farming
- ▶ Farmers allocate labor and capital b/w existing land, or cultivate new land
- ▶ Let γ denote input share allocated to cultivating new land
- ▶ Production function for new and existing land are $f(\cdot)$ and $g(\cdot)$
- ▶ Exogenous productivity shock, Ω , improves productivity of existing land more

$$\frac{\partial g(l)}{\partial \Omega} \geq \frac{\partial f(l)}{\partial \Omega} \geq 0 \quad \forall l$$

Conceptual framework

- ▶ Farmers are factor market constrained (Conning and Udry, 2007)
 - ▶ Cannot hire inputs beyond initial endowment
- ▶ Total deforestation given by new land cleared for agriculture:

$$D^* = n^* \cdot \gamma^*$$

- ▶ Differentiating with respect to Ω gives:

$$\frac{\partial D^*}{\partial \Omega} = \underbrace{n^* \frac{\partial \gamma^*}{\partial \Omega}}_{\leq 0} + \underbrace{\gamma^* \frac{\partial n^*}{\partial \Omega}}_{\leq 0}$$

- ▶ What does each term represent and what is the sign?

Abman et al. (2023): Agricultural Productivity and Deforestation

- ▶ **Goal:** To test $\frac{\partial D^*}{\partial \Omega}$, we need random variation in Ω
- ▶ **Question:** what is the impact of ag productivity on deforestation?
- ▶ **Context:** Large scale ag extension program in Uganda (2008-2013)
- ▶ **Design:** Leverage Spatial discontinuity in village eligibility
- ▶ **Results:** improvement in agricultural productivity; reduction in deforestation

Background: Why Uganda?

- ▶ Forest cover shrunk from 24% to 9% from 1990-2015
- ▶ Most forest land is **privately owned**
 - ▶ Land owners allowed to covert forest (Land Act, 2010)
 - ▶ Cannot use state powers to limit deforestation
- ▶ Need an indirect mechanism to incentivize conservation

Background: BRAC Extension Program

- ▶ Launched in 2008
- ▶ Aims to extend modern cultivation techniques to smallholders
- ▶ Two step treatment:
 - ▶ Training: “model farmers’ ’ trained in modern cropping, irrigation, weeding, pest control. Then, set up demo plot and train others
 - ▶ Extension: Promoters selected to sell subsidized HYV seeds in their villages
- ▶ **Key feature: Program limited to villages within 6km of each BRAC branch**

- ▶ Deforestation from GFC (Hansen et al., 2013)
 - ▶ Pro: high resolution (30m)
 - ▶ Con: only captures loss not gain
- ▶ Lat/lon of villages within 12km of BRAC office (N=807 villages)
- ▶ Attribute forest pixel to village if w/n 400m of village centroid
 - ▶ Rationale: median household distance to village center = 400m
 - ▶ Problems?
- ▶ BRAC agricultural survey on 7,781 households (451 villages) in 2011
 - ▶ Detailed cropping practices from last two seasons

Empirical Design: Spatial Regression Discontinuity

- ▶ Estimate ITT since we do not know true treatment designation
- ▶ Local linear regression to estimate left and right side of discontinuity

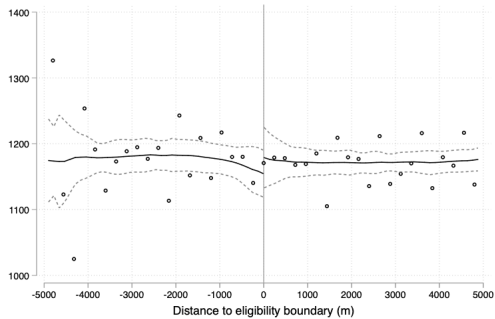
$$\beta = \lim_{z \uparrow 0} E[Y|z_i = z] - \lim_{z \downarrow 0} E[Y|z_i = z]$$

- ▶ where running variable, z_i , is distance of village from 6km cutoff
 - ▶ $z \leq 0$ means village had access to extension
- ▶ For estimation, use 2km bandwidth
 - ▶ Higher regression weights on points nearer to cutoff (triangle kernel)

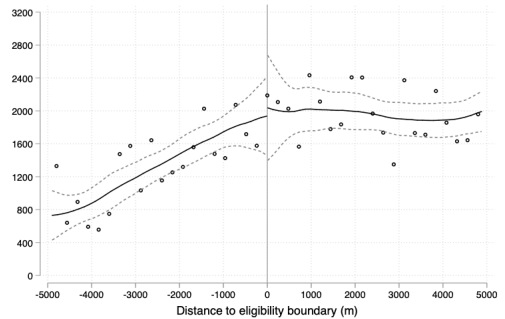
Identification Assumptions?

Identification Assumptions

(a) Elevation (meters)

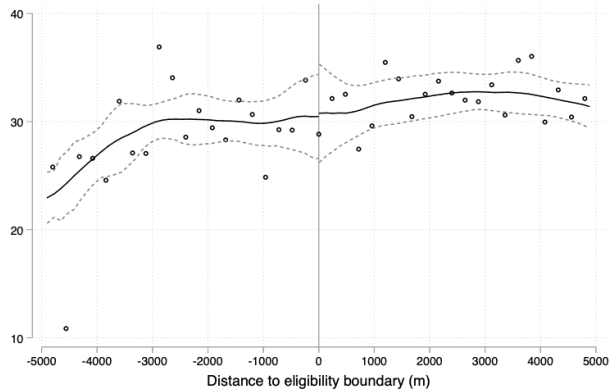


(b) Distance to nearest road (meters)



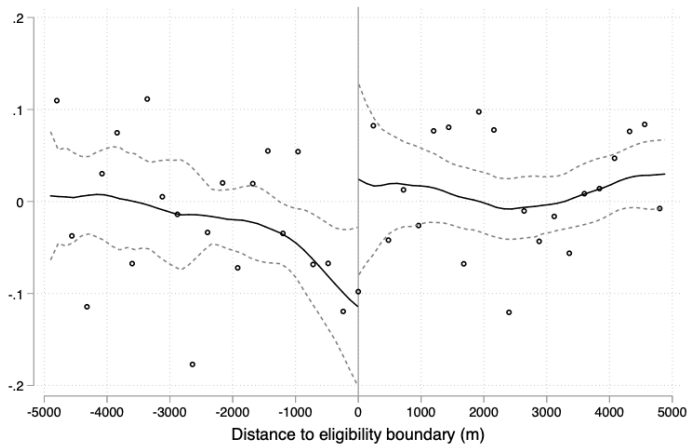
Identification Assumptions

(c) Baseline treecover



Results: 14% less deforestation in treated villages

(a) Residualized Forest Loss - Treatment period



Mechanisms: Intensification

Outcome	(1) Manure Use	(2) Intercrop	(3) Crop Rotation	(4) Irrigation	(5) Weeding	(6) Fertilizer Use	(7) HYV Seeds	(8) Perennial crops
Program Eligible	0.0977*** (0.0249)	0.0590* (0.0308)	0.0737*** (0.0251)	0.0326*** (0.00829)	0.0644** (0.0310)	-0.0171 (0.0161)	-0.0440 (0.0323)	-0.0264 (0.0351)
Obs	2912	2912	2912	2912	2912	2912	2912	2912
Control mean	0.0731	0.796	0.797	0.0266	0.693	0.0725	0.356	0.351

- ▶ Techniques to address nutrient depletion **on existing land**
- ▶ No evidence of increased fertilizer, pesticides, or HYV seed adoption

Intensification vs. Extensification

Outcome	(1) Revenue per acre	(2) IHS Cultiv ag area
Program Eligible	0.325* (0.189)	0.0262 (0.0514)
Obs	2843	2907
Control mean	11.59	1.414

- ▶ Revenue/acre is a proxy for yields
- ▶ No extensification → binding constraint on land clearing

- ▶ Empirical issues
- ▶ Alternative explanations
- ▶ External validity

Back to the Question

How does development affect the environment?

- ▶ Income
 - ▶ Land intensive consumption
 - ▶ Energy

- ▶ Access to capital
 - ▶ Technology
 - ▶ **Infrastructure**

Infrastructure and the Environment

- ▶ Roads can increase deforestation by:
 - ▶ ↓ MC (transport costs) of forest products, increasing supply
 - ▶ ↑ market access for agricultural products, causing extensification
 - ▶ ↑ land value, leading to agglomeration effects

- ▶ Roads can increase forest cover by:
 - ▶ ↑ access to forest product substitutes (i.e. clean fuel vs. firewood)
 - ▶ ↑ access to external labor/capital markets, lowering need for land-clearing

- ▶ Net effect is ambiguous

What about Infrastructure and Air Pollution?

- ▶ Lets build a conceptual framework
- ▶ Talk to your neighbor for 10 mins

Garg et al. (2023): Rural Roads, Labor Exits, and Crop Fires

- ▶ **Question:** what is the impact of roads on air quality?
 - ▶ what is the role of labor reallocation as a mechanism?
- ▶ **Context:** India builds last-mile roads using population cutoff
- ▶ **Design:** Regression discontinuity with crop fires as outcome
- ▶ **Results:** Roads $\uparrow$ crop fires and PM2.5 levels in a village
- ▶ **Mechanism:** Roads $\rightarrow$ labor exit $\rightarrow$ higher wages $\rightarrow$ labor-saving technology

Background

- ▶ Labor is misallocated in agriculture in developing countries
 - ▶ MP_{labor} lower in agriculture than other sectors
- ▶ Sources of misallocation is a big question in dev econ
- ▶ Many policies aimed at reducing misallocation
 - ▶ e.g. road-building (improve market access)
- ▶ Do these have externalities?
 - ▶ **innovation**: where do the externalities come from?

Context: India

- ▶ 14 of the 20 most polluted cities in the world are in India
- ▶ Important pollution source: crop burning
- ▶ Pros:
 - ▶ Clear rice straw residue & undergrowth b/w cropping seasons
 - ▶ Cheaper than labor/capital alternative
- ▶ Cons:
 - ▶ Carries civil/criminal penalty (weakly enforced)
 - ▶ Reduces soil nutrients and decreases ag productivity
 - ▶ Damages human health
- ▶ Roads may facilitate agricultural labor loss
 - ▶ Will farmers turn to crop burning, despite the costs?

- ▶ PMGSY aims to provide all-weather roads to all unconnected villages in India
- ▶ Prioritize larger villages first
 - ▶ Connect villages with population > 1000 by 2003
 - ▶ Connect villages with population > 500 by 2007
 - ▶ Connect villages with population > 250 afterwards
- ▶ State-by-state basis
 - ▶ If connected all large villages, can proceed to smaller ones immediately

- ▶ Roads: SHRUG database
 - ▶ N=11,151 villages w/o paved road at baseline (2001)
 - ▶ Main variable: road construction date in village
- ▶ Fires: NASA EODIS
 - ▶ # fires w/n 10km of village polygon
- ▶ Pollution: Van donkelaar et al. (2016) measure PM2.5 at 10km resolution
 - ▶ Black Carbon and Organic Carbon from biomass/non-biomass sources
- ▶ Infant mortality: NFHS-IV (2015-16)
 - ▶ child-birthyear panel from 2001-2013
 - ▶ Identify survey clusters (villages) within 50km of each PMGSY village

Empirical Strategy: Fuzzy RDD

First Stage:

$$\begin{aligned} Road_{vdst} = & \gamma_0 + \gamma_1 1(pop_{vds} \geq T) + \gamma_2 (pop_{vds} - T) \\ & + \gamma_3 (pop_{vds} - T) \times 1(pop_{vds} \geq T) + \theta X_{vds} + \mu_{d,h} + \rho_t + \epsilon_{vdst} \end{aligned}$$

Second Stage:

$$\begin{aligned} Y_{vdst} = & \beta_0 + \beta_1 Roads_{vdst} + \beta_2 (pop_{vds} - T) \\ & + \beta_3 (pop_{vds} - T) \times 1(pop_{vds} \geq T) + \delta X_{vds} + \eta_{d,h} + \omega_t + \epsilon_{vdst} \end{aligned}$$

- ▶ $Road_{vdst} = 1$ if village v received road by 2013
- ▶ Y_{vdst} is outcome (fires, pollution, mortality)
- ▶ pop_{vds} is village population in 2001
- ▶ T is treatment threshold (500 or 1000, depending on state)

Covariate Balance

Table 1: Main analysis sample: summary statistics, and balance and falsification tests

Variable	Full sample	Below threshold	Over threshold	Difference of means	p-value on difference	RD estimate	p-value on RD estimate
Primary school	0.959	0.955	0.964	0.01	0.02	-0.018	0.59
Medical center	0.166	0.155	0.177	0.02	0.00	-0.097	0.14
Electrified	0.430	0.414	0.447	0.03	0.00	-0.014	0.87
Distance from nearest town (km)	26.490	26.379	26.613	0.23	0.58	-3.426	0.34
Land irrigated (share)	0.281	0.276	0.287	0.01	0.05	-0.025	0.59
Ln land area	5.151	5.093	5.215	0.12	0.00	-0.103	0.33
Literate (share)	0.457	0.454	0.461	0.01	0.01	-0.012	0.62
Scheduled caste (share)	0.143	0.141	0.145	0.00	0.24	-0.020	0.52
Land ownership (share)	0.733	0.733	0.732	-0.00	0.75	0.013	0.72
Subsistence ag (share)	0.435	0.438	0.432	-0.01	0.25	0.024	0.58
HH income > INR 250 (share)	0.754	0.752	0.757	0.00	0.37	-0.023	0.63
Outcomes at baseline:							
Annual fires (count)	0.685	0.699	0.670	-0.03	0.40	0.194	0.50
Annual PM2.5 ($\mu\text{g}/\text{m}^3$)	35.886	35.821	35.958	0.14	0.41	-0.029	0.94
Downwind infant mortality (0/1)	0.067	0.068	0.067	-0.00	0.28	0.002	0.63
Other directions infant mortality (0/1)	0.072	0.072	0.071	-0.00	0.23	-0.002	0.72
Downwind PM 2.5 ($\mu\text{g}/\text{m}^3$)	36.038	35.879	36.252	0.37	0.04	0.431	0.36
Other directions PM 2.5 ($\mu\text{g}/\text{m}^3$)	39.254	38.646	40.237	1.59	0.00	0.901	0.15
N	11151	5850	5292				

Results

	New road	Annual fire activity		Annual average PM 2.5	
	(1) <i>1st stage</i>	(2) RF	(3) IV	(4) RF	(5) IV
Above threshold pop.	0.230*** (0.017)	0.366** (0.152)		0.106** (0.045)	
Road built			1.567** (0.689)		0.470** (0.207)
N	133,788	133,788	133,788	133,788	133,788
Control group mean	0.10	2.77	2.77	42.69	42.69

- Note: optimal bandwidth = +/- 84 around population threshold
- Road access increases annual crop fires by 60%
- Road access increase PM2.5 level by $0.5 \mu\text{g}/\text{m}^3$ (1.1%)

Does pollution result also capture increased driving on new roads?

Panel A: Winter harvest and post-harvest months							
	Fires	All sources		Biomass burning		Other sources	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		BC	OC	BC	OC	BC	OC
Road built	1.051*** (0.391)	0.285** (0.130)	3.146* (1.777)	0.320*** (0.103)	2.791** (1.180)	-0.010 (0.009)	-0.010 (0.009)
N	133,788	133,788	133,788	133,788	133,788	133,788	133,788
Control group mean	1.45	30.47	120.81	1.17	13.46	29.30	107.35
Panel B: Rest of the year							
	Fires	All sources		Biomass burning		Other sources	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		BC	OC	BC	OC	BC	OC
Road built	0.276 (0.438)	-0.219 (0.351)	-0.115 (3.120)	0.192 (0.246)	0.886 (2.680)	-0.010 (0.009)	-0.010 (0.009)
N	133,788	133,788	133,788	133,788	133,788	133,788	133,788
Control group mean	1.27	30.83	124.70	1.53	17.34	29.31	107.36

Mechanism: Roads increase ag wage rate

- ▶ Data: REDS Survey (1999 and 2006)
- ▶ Design: Diff-in-Diff (221 villages, 100 districts, 17 states)
 - ▶ Treat = 1 if v received road b/w 1999-2006; Post=1 in 2006

$$Y_{vt} = \alpha + \beta_1(Treat_v \times Post_t) + \gamma_v + \theta_t + \epsilon_{vt}$$

	Wage rate			Log wage		
	(1) Male	(2) Female	(3) Average	(4) Male	(5) Female	(6) Average
Treat X Post	0.722* (0.420)	0.918*** (0.349)	0.856** (0.352)	0.012* (0.007)	0.020** (0.008)	0.016** (0.007)
N	442	442	442	442	442	442
Control group mean	62.99	47.65	55.25	62.99	47.65	55.25
R ²	0.76	0.78	0.77	0.75	0.76	0.76

Mechanism: Roads induce labor exit

- ▶ Back to RDD design
- ▶ Data from Socioeconomic and Caste Census (2013), provided in SHRUG
- ▶ Outcome: ag. labor share in districts with high/low ag wage

	Share of labor in agriculture		Share of non-agricultural manual labor	
	(1) High rel. ag. wage	(2) Low rel. ag. wage	(3) High rel. ag. wage	(4) Low rel. ag. wage
Road built	-0.029 (0.046)	-0.245*** (0.093)	0.030 (0.046)	0.210** (0.092)
N	5,402	5,483	5,402	5,483
Control group mean	0.49	0.46	0.45	0.46

- ▶ More labor exit in districts with lower baseline ag wages

Mechanism: places with more labor exit burn more biomass

	High rel. ag. wage		Low rel. ag. wage	
	(1)	(2)	(3)	(4)
	Fires	PM 2.5	Fires	PM 2.5
Road built	0.778	0.378*	3.195**	0.617*
	(0.721)	(0.220)	(1.439)	(0.367)
N	62,880	62,880	67,740	67,740
Control group mean	2.68	45.38	2.88	40.09

- ▶ Fire/pollution impact greater in places with lower baseline ag wage
- ▶ Road generates more labor exit in these places

- ▶ Do you believe the story?
- ▶ Alternative explanations

Next Time

- ▶ Presentations
- ▶ How does the environment affect development?
 - ▶ **Health**
 - ▶ Productivity